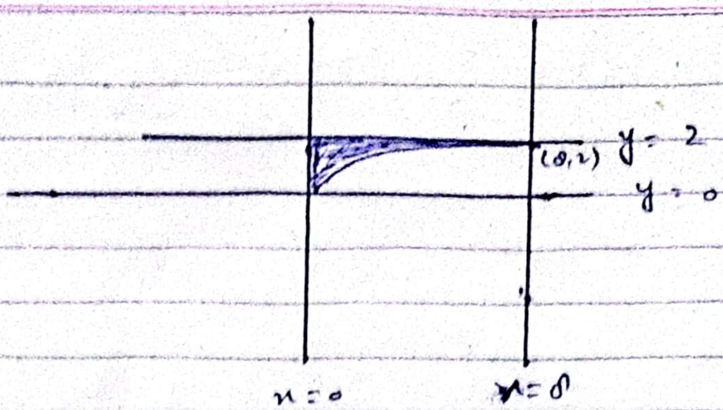


$$\int_0^2 \int_0 \frac{y^3}{1+y^4}\, dx\, dy$$

Ex # 15.4 Double Integrals in Polar form

$$\iint_R e^{x^2+y^2}\, dA = \int_{-1}^{1}\int_0^{\sqrt{1-x^2}} e^{x^2+y^2}\, dy\, dx$$

(V.C)

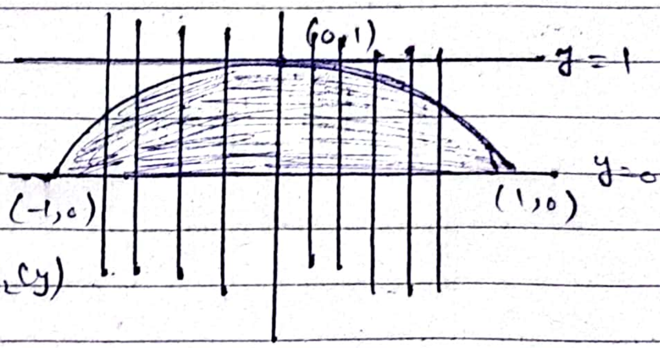
R: Semi circular region

$a \le x \le b$

$g_1(x) \le y \le g_2(x)$

$y = 0$

$y = \sqrt{1-x^2}$



(H.C)

$c \le y \le d \qquad h_1(y) \le x \le h_2(y)$

$0 \le y \le 1 \qquad -\sqrt{1-y^2} \le x \le \sqrt{1-y^2}$

$y^2 = 1 - x^2$

$x = \pm\sqrt{1-y^2}$

$\int \sqrt{a^2+x^2}\, dx$

$x = a\tan\theta$

$dx = a\sec^2\theta\, d\theta$

Jacobian of the transformation

Jacobian of transformation

$x = r\cos\theta \qquad r = \sqrt{x^2+y^2}$

$y = r\sin\theta \qquad \tan\theta = y/x$

L from origin cutting through the region R, in the direction of increasing r.
Mark r-values where ray L enters and leaves the region. These are r-Limits. They usually depends on θ, that L makes with +ve x-axis

$$= \int_0^{\pi}\int_0^1 e^{r^2} r \, dr \, d\theta$$

$$= \frac{1}{2}\int_0^{\pi}\int_0^1 2e^{r^2} r \, dr \, d\theta = \frac{1}{2}\int_0^{\pi} e^{r^2}\Big|_0^1 d\theta$$

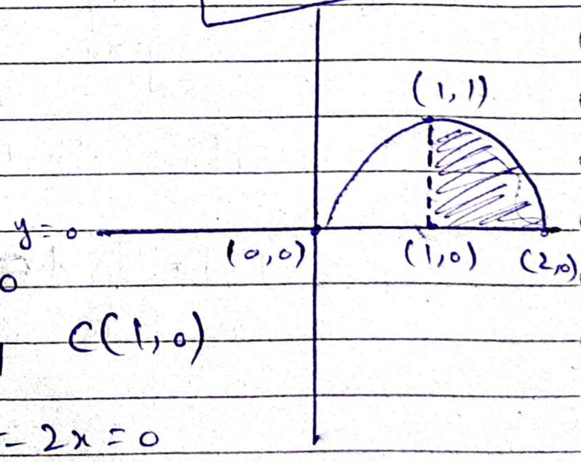
$$= \frac{1}{2}\int_0^{\pi} (e-1)\, d\theta$$

$$= \frac{\pi}{2}(e-1)$$

Q 22:

$$\int_1^2 \int_0^{\sqrt{2x-x^2}} \frac{dy\,dx}{(x^2+y^2)^2}$$

$$\boxed{\theta = \tan^{-1}\left(\frac{y}{x}\right)}$$



$y=0$

$y=\sqrt{2x-x^2} \Rightarrow y^2+x^2-2x=0$

$y^2+(x-1)^2=1 \quad C(1,0)$

$x=1$

$r\cos\theta = 1$

$\boxed{r=\sec\theta}$

$y^2+x^2-2x=0$

$r^2-2r\cos\theta=0$

$r \neq 0$

$r=2\cos\theta.$

$$\int_0^{\pi/4}\int_{\sec}^{2\cos\theta} \frac{r\,dr\,d\theta}{r^4}$$

$$\int_0^{\pi/4}\int_{\sec\theta}^{2\cos\theta} r^{-3}\,dr\,d\theta = -\frac{1}{2}\int_0^{\pi/4} r^{-2}\Big|_{\sec\theta}^{2\cos\theta}\,d\theta$$

$$= -\frac{1}{2}\int_0^{\pi/4}\left[\frac{1}{4\cos^2\theta} - \cos^2\theta\right]d\theta$$

$$= -\frac{1}{2}\int_0^{\pi/4}\left[\frac{1-4\cos^4\theta}{4\cos^2\theta}\right]d\theta$$

Ex # 15. Triple Integrals In Rectangular Co-ordinates.

$w = F(x,y,z) \rightarrow$ D: Solid Regions $\mathbb{R}^3$
↳ Combination of surfaces

(x, y, z, w)

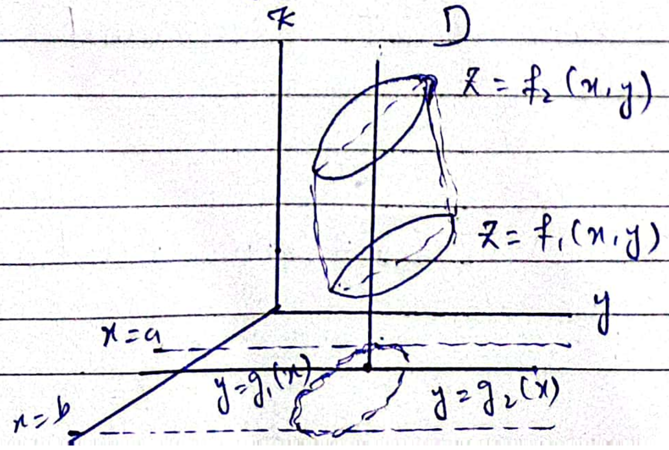
$$\boxed{\begin{array}{l} z = f(x,y) \\ (x, y, f(x,y)) \end{array}}$$

$$\iiint_D F(x,y,z)\,dV$$

$$\left.\begin{array}{l} dV = dx\,dy\,dz \\ \quad\;\; dx\,dz\,dy \end{array}\right] \text{Image in } yz\text{-plane}$$

$$\left.\begin{array}{l} = dy\,dz\,dx \\ \quad\; dy\,dx\,dz \end{array}\right] \quad xz\text{-plane}$$

$$\left.\begin{array}{l} = dz\,dx\,dy \\ \quad\; dz\,dy\,dx \end{array}\right] \quad xy\text{-plane}$$



$$\int_a^b \int_{g(x)}^{g_1(x)} \int_{f_1(x,y)}^{f_1(x,y)} F(x,y,z)\, dz\, dy\, dx.$$

Finding Limits of Integration in the Order $dz\,dy\,dx$:

1) Sketch: Sketch the region D (if Possible) along with its shadow R (vertical Projection) in xy-plane. Label the bending surfaces of D and bounding curves of R.

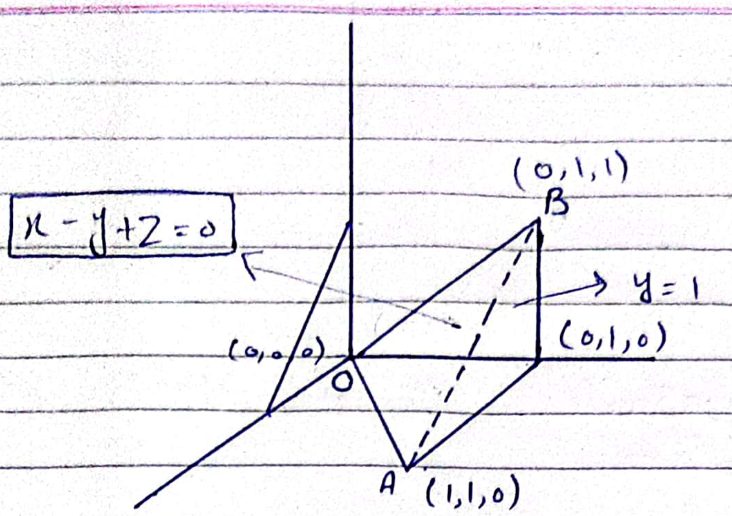
2) Z-Limits of Integration: Draw a line M passing through the (x,y) in R parallel z-axis. As z increases M enters D at $z = f_1(x,y)$ and leaves at $z = f_2(x,y)$. These are z limits of integrals.

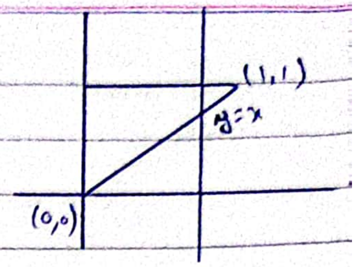
3) Y-Limits: Draw a line L through the point (x,y) parallel to y-axis. As y increases the line L enters R at $y = g_1(x)$ and Leaves at $y = g_2(x)$.

4) X-Limits: Choose all lines parallel to y-axis that surround the region R.

Example: 2 & 3

$(0,0,0)$, $(1,1,0)$, $(0,1,0)$, $(0,1,1)$





$\vec{OA} = \hat{i} + \hat{j}$

$\vec{OB} = \hat{j} + \hat{k}$

$\vec{n} = \vec{OA} \times \vec{OB} = \hat{i} - \hat{j} + \hat{k}$

$P(x_0, y_0, z_0)$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

↑ general Equation of plane

↳ $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$ (normal to the plane)

↳ general Equation of plane

$$1(x-0) - 1(y-0) + 1(z-0) = 0$$

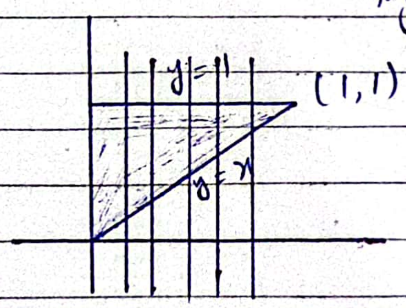
$$x - y + z = 0$$

$$\int_0^1 \int_x^1 \int_0^{y-x} F(x,y,z)\,dz\,dy\,dx$$

↳ discard V_D.

xy-plane

Shadow:

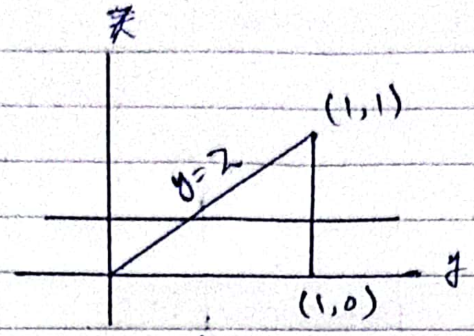


$$\int_0^1 \int_z^1 \int_0^{y-z} dx\,dy\,dz$$

↳ dz dy

↳ x limits depends on y & z / Projection on yz-plane

Shadow:

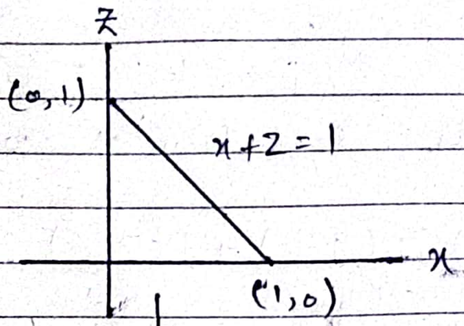


A line parallel to x starts from projection plane.

$$\int_0^1 \int_0^{1-z} \int_{x+z}^1 dy\,dx\,dz$$

$$\int_0^1 \int_0^{1-x} \int_{x+z}^1 dy\,dz\,dx$$

Shadow:

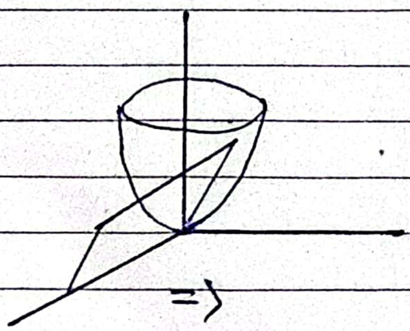


Projection down to y.

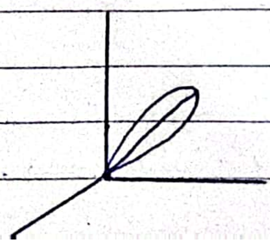
Q #6

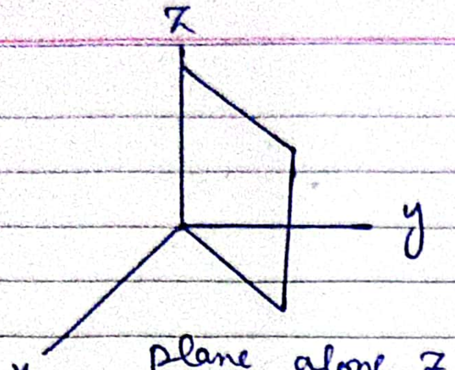
D: $z = x^2 + y^2$ (Paraboloid)

$z = 2y$

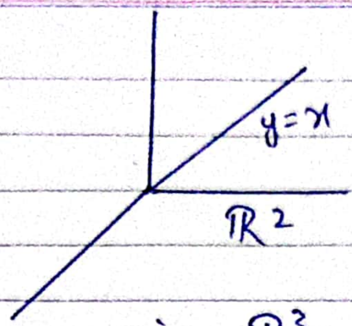


$\Rightarrow$

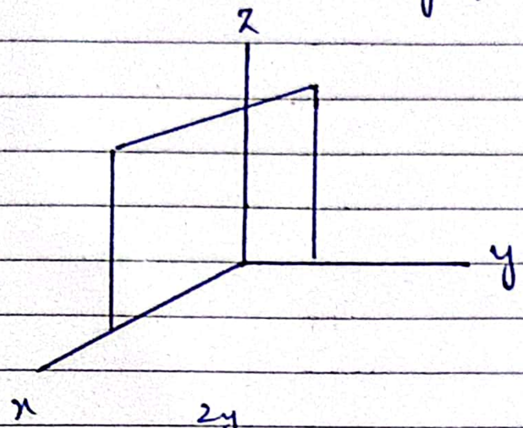




plane along z

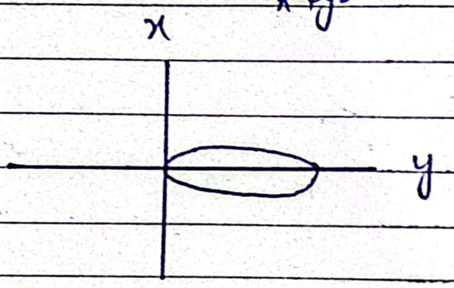


$y = x$ in $\mathbb{R}^3$ is plane



$$\int\int\int_{x^2+y^2}^{2y} dz\,dx\,dy$$

$dy\,dx$



Intersection

$$x^2 + y^2 = 2y$$

$$x^2 + (y-1)^2 = 1$$